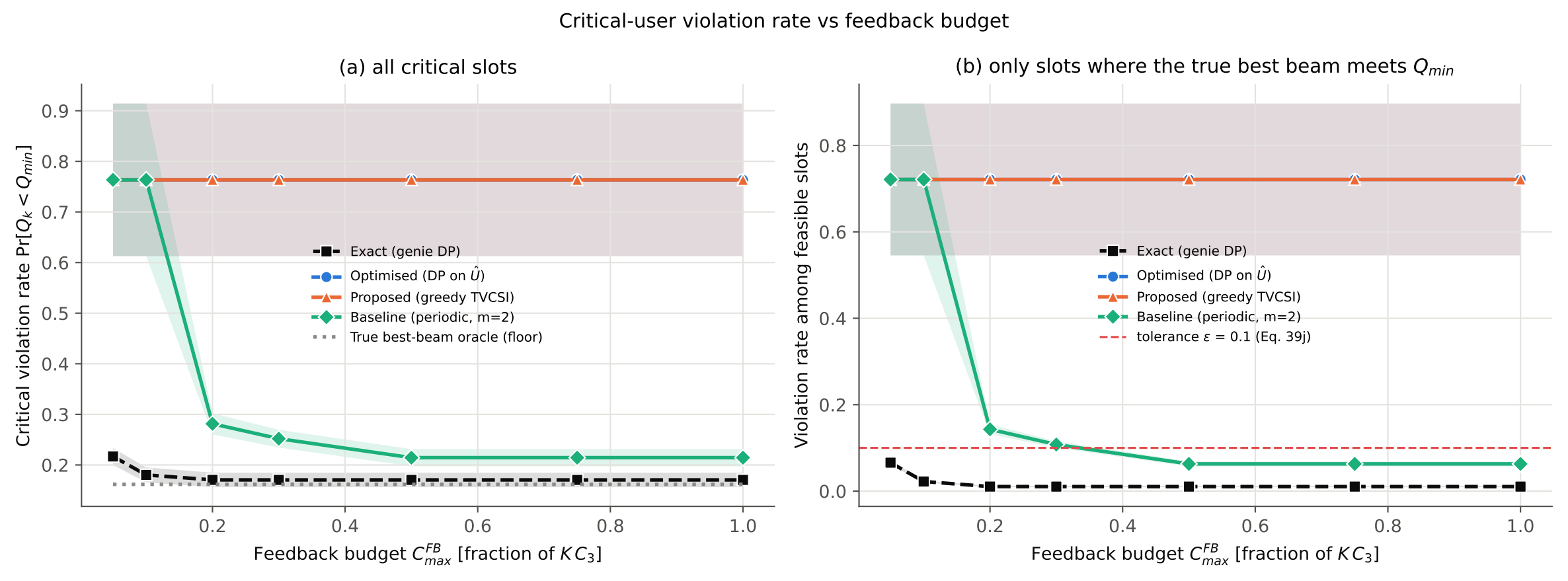
Critical-user violation rate vs feedback budget
(a) all critical slots
(b) only slots where the true best beam meets Q_{min}
Critical violation rate $\Pr[Q_k < Q_{min}]$
Violation rate among feasible slots
Feedback budget C^{FB}_{max} [fraction of KC_3]
Feedback budget C^{FB}_{max} [fraction of KC_3]
Exact (genie DP)
Optimised (DP on $\hat{U}$)
Proposed (greedy TVCSI)
Baseline (periodic, m=2)
True best-beam oracle (floor)
tolerance ε = 0.1 (Eq. 39j)
DUMMY MODELS
mean over 10 test seeds, band = 95% t-interval; 16.2% of critical slots are infeasible even with the true best beam